# KANCHANI RAJYALAKSHMI

Software Engineer

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## PROFESSIONAL SUMMARY

B.Tech Computer Science student (Class of 2027) with software engineering internship experience at UZO, a cloud-native platform startup. Strong programming fundamentals in Python, SQL and JavaScript, with hands-on experience building and testing REST APIs with FastAPI and reviewing code through GitHub pull requests. Builds AI applications using LLM APIs, retrieval-augmented generation (RAG), embeddings and tool calling, and evaluates output quality. Detail-oriented and analytical, with clear written communication for explaining technical decisions; ServiceNow certified.

## CORE SKILLS

**Languages**: Python, SQL, JavaScript, HTML, CSS
**Frameworks & Tools**: FastAPI, React, REST APIs, VS Code
**Development Practices**: Code Reviews, API Testing (Postman), Agile Workflows, Git, GitHub
**AI / ML**: LLM APIs, Prompt Engineering, RAG, Embeddings, Vector Search, Tool Calling, Evaluation
**Databases**: PostgreSQL, MongoDB, SQL
**Core CS**: OOPs, DBMS, Operating Systems, Data Structures & Algorithms

## PROFESSIONAL EXPERIENCE

**Software Engineering Intern** – UZO (Startup) *Apr 2026 – Jun 2026*

- Contributed to backend and product features at UZO, a cloud-native platform startup, as part of an agile engineering team.
- Built and tested REST API endpoints in Python and FastAPI, validating requests, responses and edge cases with Postman to ensure correctness and reliability.
- Collaborated through Git and GitHub with pull requests and code reviews, providing structured feedback on teammates' code within agile sprint workflows.

## PROJECTS

### AI Healthcare Assistant

- Built a RAG-based healthcare assistant that answers questions from approved medical information with source citations.
- Implemented authentication and role-based access controls using FastAPI and PostgreSQL to protect user data.
- Added safe escalation that detects medical-emergency queries and directs users to emergency services instead of answering.
- Evaluated answer quality and deployed the application.

### Research Assistant

- Created an agent workflow that searches approved sources, summarizes evidence and produces structured reports with cited references.
- Used LLM tool calling to orchestrate search, extraction and summarization steps, with evaluation checks on report accuracy.

### PDF Question-Answering Assistant

- Built an assistant that lets users upload PDFs and ask questions, returning answers with page-level source references.
- Implemented text extraction, chunking, embedding generation and vector search to retrieve the most relevant passages.

### AI Chatbot

- Developed a chat interface connected to an LLM API with streamed responses, error handling and saved conversation history.

### GeneWeaver: GPU-Accelerated CRISPR Alignment Engine

- Developed a GPU-accelerated CRISPR off-target alignment engine to process large-scale genomic datasets for DNA sequence analysis.
- Engineered CUDA JIT kernels and optimized GPU memory management to accelerate sequence alignment through parallel execution.
- Designed a Dask pipeline for multi-GPU scheduling and a Textual dashboard to monitor GPU usage and alignment progress.

### AI Hospital Appointment Booking

- Developed an AI-powered healthcare platform with face-recognition login and JWT-based access control.
- Integrated medical report analysis, doctor appointment booking, medicine reminders and emergency services.

## EDUCATION

**B.Tech, Computer Science and Engineering** *2027 | Kavali, Andhra Pradesh | CGPA: 86*
*PBR Visvodaya Engineering College*

## CERTIFICATIONS

**ServiceNow Certified System Administrator | ServiceNow Certified Application Developer**
**freeCodeCamp Scientific Computing with Python**